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# AMPERE DEVELOPMENT

Power Testing & Load Bank Solutions

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## Load Bank Setup & Test Procedures

A Practical Guide for Safe and Effective Power System Testing

This guide outlines recommended practices for the safe setup, operation, and documentation of load bank tests. It is intended for qualified electrical personnel performing commissioning, acceptance, or periodic testing of generators, UPS systems, batteries, data-center power infrastructure, and medium-voltage equipment. Always follow site-specific safety rules, manufacturer instructions, and applicable codes (NFPA 110, IEEE, local AHJ requirements).

**Ampere Development provides the equipment, expertise, and support to execute these procedures correctly the first time.**

### 1. Safety First

Load bank testing involves high currents, significant heat, rotating equipment, and potential arc-flash hazards. Only qualified, trained personnel should perform these procedures.

#### Minimum Recommended PPE

- Safety glasses or goggles, gloves rated for the voltage, steel-toe boots, high-visibility vest
- Hearing protection near operating load banks (fan noise)
- Arc-flash PPE appropriate to the system voltage and available fault current (especially medium-voltage work)
- Flame-resistant clothing when required by site policy or calculated incident energy

#### Key Safety Rules

- Never bypass interlocks, airflow switches, or protective devices on the load bank.
- Keep clear of exhaust airflow — discharge temperatures can exceed 200 °F (93 °C).
- Maintain a safety perimeter; restrict access to authorized personnel only.
- Confirm proper grounding of both the load bank and the unit under test.
- Have a clear emergency shutdown plan and know the location of E-stops.

### 2. Pre-Test Planning & Preparation

Successful testing begins with planning. Confirm the following before mobilizing equipment:

- Nameplate ratings of the unit under test (kW, kVA, voltage, frequency, power factor)
- Required test profile (acceptance, NFPA 110 annual, commissioning, battery discharge, etc.)
- Available connection points, cable lengths, and ampacity requirements
- Site access, clearance, ventilation, and heat rejection path for the load bank
- Fuel, coolant, and battery status of the generator or UPS under test
- Any paralleling, transfer-switch, or building-load coordination requirements
- Data-logging needs and acceptance criteria agreed with the owner or AHJ

### 3. Pre-Test Inspection

#### Unit Under Test (Generator / UPS / Battery)

- Fuel level adequate for planned duration plus margin; oil and coolant levels within specification
- Battery voltage, connections, and electrolyte (if applicable) verified
- Exhaust system clear; no visible wet-stacking residue or leaks
- Control panel free of active fault codes; belts, hoses, and wiring inspected

- Adequate ventilation and cooling air path available for both unit and load bank

### Load Bank & Cables

- Visual inspection for damage, blocked airflow, or missing guards
- All load switches in the OFF position before connecting
- Cables correctly sized, undamaged, and terminated securely
- Voltage selector (if present) set to match the unit under test
- Control power / auxiliary supply available and correctly configured

## 4. Load Bank Setup & Connection

Position the load bank with adequate clearance for intake and exhaust airflow. Never obstruct the discharge path. For outdoor units, consider wind direction and proximity to buildings or personnel.

### Connection Best Practices

- Match voltage, phase sequence, and frequency of the unit under test
- Use properly rated, labeled cables; torque connections to manufacturer specifications
- Prefer dedicated load-bank connection points or paralleling buses downstream of overcurrent protection
- Do not disconnect the generator from building loads solely to connect the load bank if emergency power must remain available
- Verify phase rotation and grounding before energizing
- Confirm load-bank cooling fans and control circuits operate correctly at zero load

*Note: For medium-voltage or complex systems, Ampere technicians can provide on-site setup support and connection verification.*

## 5. Start-Up and Warm-Up

Start the unit under test with no load applied. Allow it to reach normal operating temperature, stable voltage, and stable frequency (typically 5–15 minutes depending on ambient conditions and equipment size). Listen for abnormal noises, check for leaks, and verify baseline readings before applying any load.

## 6. Applying Load – Step Loading Procedure

Apply load gradually in controlled steps. This reveals problems at lower stress levels and protects both the unit under test and the load bank. Never apply full load in a single step unless specifically required by a manufacturer procedure.

### Typical Step Sequence (General / Commissioning)

| Step         | Load Level | Typical Duration | Primary Purpose                             |
|--------------|------------|------------------|---------------------------------------------|
| 1            | 25–30 %    | 10–15 min        | Warm-up, baseline readings, initial checks  |
| 2            | 50 %       | 15–30 min        | Mid-range stability, thermal response       |
| 3            | 75 %       | 30 min           | Higher thermal stress, cooling verification |
| 4            | 100 %      | 60–120 min       | Full capacity, heat soak, acceptance        |
| 5 (optional) | 110 %      | 5–10 min         | Overload capability (if specified)          |

### NFPA 110 Annual Load Test (2025 Protocol – Reference)

For emergency and standby power systems, the current NFPA 110 annual test protocol is typically: **50 % of nameplate kW for 30 continuous minutes**, followed by **75 % of nameplate kW for 60 continuous minutes**. Confirm the exact requirements applicable to your facility and AHJ.

## 7. Parameters to Monitor During the Test

Record readings at regular intervals (every 5–15 minutes is common). Watch for trends, not just single values.

- Voltage (all phases) and voltage balance
- Frequency / engine speed stability
- Current (per phase) and power (kW / kVA)
- Power factor (especially when using reactive or combined load banks)
- Oil pressure, coolant temperature, exhaust gas temperature
- Load-bank airflow, element temperature indicators, and any alarms

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- Unusual noise, vibration, smoke, or fluid leaks
  - For UPS / battery tests: battery string voltage, individual cell voltages, runtime

**Stop the test immediately if voltage or frequency drifts outside manufacturer limits, temperatures rise uncontrollably, or any safety device activates.**

## 8. Unloading, Cool-Down & Demobilization

Remove load gradually in reverse steps. Allow the unit under test to run unloaded for a cool-down period per manufacturer recommendations (often 5–15 minutes). Then shut down, perform a post-test visual inspection, and disconnect the load bank only after confirming the system is de-energized and safe.

- Return transfer switches and breakers to normal operating positions
- Inspect for leaks, overheating marks, or debris
- Secure and store cables; leave the load bank ready for transport

## 9. Documentation & Reporting

A test that is not properly documented is often treated as a test that never occurred. Capture at minimum:

- Date, time, location, equipment identification, and personnel involved
- Load steps, durations, and measured values (voltage, frequency, kW, temperatures, etc.)
- Any anomalies, alarms, or corrective actions taken
- Final pass/fail determination against the agreed acceptance criteria
- Photographs or data-logger exports as supporting evidence

## 10. Special Applications Notes

**Data Centers / AI Power:** Server-simulating or rack-mounted load banks provide realistic thermal and electrical profiles. Coordinate with cooling-system testing (IST) and ensure heat rejection does not disrupt other operations.

**Medium-Voltage Systems:** Require specialized equipment, higher PPE levels, and often utility or owner coordination. Ampere can supply medium-voltage load banks and experienced technicians.

**UPS & Battery Discharge:** Use calibrated resistive or programmable electronic load banks. Monitor cell voltages closely and have a clear abort criterion for low-voltage cutoff.

**Reactive / Combined Testing:** When power-factor performance must be verified, use resistive-reactive load banks. This more accurately stresses both the engine and the alternator.

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### Need Equipment or On-Site Support?

Ampere Development supplies resistive, reactive, DC, electronic, medium-voltage, and specialty load banks nationwide — with expert setup assistance available.

**888-345-LOAD · [ampere.dev](https://ampere.dev) · [sales@ampere.dev](mailto:sales@ampere.dev)**